HANZLA AHMAD

Full-Stack & Systems Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack & Systems Software Engineer with 4+ years of hands-on experience building production web platforms, deterministic Linux background daemons, and Generative AI agent systems. Proven expertise engineering type-safe architectures with Next.js 15, React 19, and TypeScript, building low-level Unix services with WebSockets and IPC, and deploying tamper-resistant database platforms with PostgreSQL and SQLite. Experienced in managing the entire software lifecycle from requirements synthesis to containerized Linux VPS deployment.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C++, Java, Dart, SQL, Bash / Shell Scripting

Frontend & Web: Next.js 15 (App Router, Server Actions, RSC), React 19, Tailwind CSS, HTML5, CSS3, Vite, Responsive UI/UX

Backend & APIs: Node.js, Express.js, RESTful APIs, WebSockets, Prisma ORM, Microservices Architecture, Unix Domain Socket IPC

Databases & Storage: PostgreSQL, SQLite (WAL mode), MongoDB, Redis, Supabase, Cloud Firestore

AI & Agentic Systems: Google Gemini API, Claude API, LLM Function Calling, RAG (Retrieval-Augmented Generation), Prompt Engineering

DevOps & Systems: Linux (Ubuntu/Debian), systemd Daemons, Docker, Nginx (Reverse Proxy), CI/CD (GitHub Actions), Git, Vercel

PROFESSIONAL EXPERIENCE

HA Entertainment | *Founder & Lead Systems Engineer*

Jun 2020 – Present | Lahore, Pakistan

- Architected and deployed full-stack digital media platforms and operational infrastructure serving active client workloads with sub-second response times.
- Designed high-performance web frontends and microservices utilizing Next.js, React, Node.js, and TypeScript with strict type safety.
- Provisioned and managed Linux VPS servers, configuring automated cron backups, Nginx reverse proxies, SSL/TLS certificates, and systemd services with 99.9% uptime.
- Directed continuous delivery (CI/CD) pipelines across Vercel and bare-metal environments, ensuring zero-downtime production releases.

MedBills MBS | *Information Technology Administrator*

Nov 2025 – Oct 2026 | Lahore, Pakistan

- Administered core enterprise IT infrastructure, network security, and portal workflows supporting medical billing operations.
- Enforced role-based access control (RBAC), end-to-end data encryption, and network firewall policies compliant with industry privacy standards.
- Automated weekly database backups, system health monitoring scripts, and credential rotation protocols, resolving 100% of reported infrastructure tickets within SLA.

Ajwa Mart Sweets & Bakers | *IT Systems & POS Specialist*

Jul 2021 – Present | Chichawatni, Pakistan

- Engineered Point-of-Sale (POS) synchronization and multi-branch retail inventory tracking across 140+ retail stock items.
- Developed custom Shopify Liquid modules, optimizing checkout flow, cart performance, and responsive product catalog browsing.
- Automated dual-shift end-of-day sales reconciliation workflows, eliminating discrepancies between register cash counts and inventory balances.

IT Beam | *MERN Stack Developer Intern*

Jul 2026 – Sep 2026 | Lahore, Pakistan

- Developed core modules of an enterprise Learning Management System (LMS) using MongoDB, Express.js, React.js, and Node.js.
- Implemented role-based dashboards (Admin, Instructor, Student), course curriculum management, and automated quiz evaluation engines.
- Built and documented RESTful API endpoints with JWT authentication and input validation, improving API test coverage.

DevelopersHub Corporation | *Software Engineering Intern*

Mar 2025 – May 2025 | Remote

- Collaborated in an agile scrum team to deliver feature enhancements and bug fixes for responsive web applications.
- Participated in weekly code reviews, Git feature-branch workflows, and automated regression testing.

FEATURED ENGINEERING PROJECTS

WA-Agent — Autonomous AI Agent Control Plane & Terminal Supervisor | *Stack: TypeScript, Node.js, Baileys WebSockets, Unix Domain Sockets, systemd*

- Engineered a deterministic remote control plane allowing terminal-based CLI agents (e.g., Claude Code) to be supervised via WhatsApp.
- Architected dual-process daemon architecture with token-authenticated Unix Domain Socket IPC for secure command dispatching.
- Implemented real-time bidirectional streaming of terminal stdout/stderr chunks with auto-reconnection and session persistence.

ReqForge — AI Requirements Engineering & PRD Platform | *Stack: Next.js, React, TypeScript, Google Gemini 2.0 API, Tailwind CSS*

- Developed an AI Studio application that transforms ambiguous product briefs into structured, IEEE 830-compliant Software Requirement Specifications (SRS/PRD).
- Designed multi-stage prompt orchestration pipelines with strict JSON schema validation and export capabilities (Markdown, PDF).

Franchise Expense Reconciliation System | *Stack: Node.js, Express.js, PostgreSQL, Prisma ORM, Tailwind CSS*

- Built an enterprise operational accounting system deployed in production at Jazz Franchise 7301 for daily petty cash and expense reconciliation.
- Structured tamper-resistant audit logs and shift closure verification, ensuring 100% financial accuracy across physical cash handoffs.

Pepsi Stock Balance Portal | *Stack: Next.js 15, React, TypeScript, Prisma ORM, SQLite / PostgreSQL*

- Built a commercial beverage distribution and inventory reconciliation platform tracking delivery truck stock vs empty crate returns.
- Implemented real-time batch reconciliation algorithms preventing inventory variance across warehouse dispatch cycles.

whatsapp-widget-daemon | *Stack: Node.js, Baileys, Unix Domain Sockets, Linux IPC*

- Developed a low-overhead headless background daemon over Baileys WebSockets providing desktop widget frameworks with live WhatsApp event data over local Unix sockets.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Riphah International University, Lahore, Pakistan | Feb 2024 – Feb 2028

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Object-Oriented Programming, Theory of Automata.

VERIFIED CERTIFICATIONS & CREDENTIALS

Google Cloud Skills Boost: Responsible AI: Applying AI Principles with Google Cloud (2026), Introduction to Large Language Models (2026), Introduction to Generative AI (2026)

Cisco Networking Academy: Python Essentials 1 (2026), Python Essentials 2 (2026), Linux Unhatched (2026), Introduction to Cybersecurity (2026), Networking Basics (2026), Computer Hardware Basics (2026), Operating Systems Basics (2026), JavaScript Essentials 1 & 2 (2026), C++ Essentials 2 (2025), Introduction to IoT (2024)

Language Proficiency: fondi Inc: CEFR B2 Independent English Proficiency Certification (2024)